Axel Gomez

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Axel Gomez AxelGomez16 Web profile
Queretaro, Mexico — Open to Relocation

Professional Summary

Algorithms Engineer with **4+ years in automotive R&D**, combining Python/C++ evaluation tools with **commercial LLM integration**. Pursuing an AI-focused M.S., with **scikit-learn/PyTorch coursework** and ongoing hybrid metaheuristic research. Builds ground-truth metrics and automated validation to make sensor performance measurable.

Professional Experience

Aumovio (formerly Continental R&D) June 2022 - Present
Algorithms Engineer *Queretaro, Mexico*

- **Radar Model Validation:** Proposed a **Python** pipeline, then refactored and migrated validation code with **GitHub Actions**, reducing a multi-engineer workflow from approximately **five working days to 50 minutes** for General Motors.
- **LiDAR Perception & Evaluation:** Built ground-truth evaluation for static-object detection and bounding-box query with **SQLite/Dash/Plotly**, making metrics available across teams and surfacing regressions through commit/pull-request checks.
- **Perception Integration:** Established two-week checks of combined algorithm performance to inform release decisions; developed **C++** visualization features and prepared customer demos.
- **Algorithm Orchestration:** Migrated legacy **C++** scheduling for parallel camera/radar algorithms and per-frame failure handling.
- **Team Enablement & Code Quality:** Set up Linux environments, trained teammates on internal tools, and helped resolve QAC findings across algorithm components under AUTOSAR/MISRA.

Postealo - Independent Software Product December 2024 - Present
Developer & Maintainer (alongside full-time employment) *Queretaro, Mexico*

- Built a **Python/LLM** workflow from bot intake to editorial preparation and immediate/scheduled publishing across five channels; a paying news organization uses it under a **monthly subscription** with ongoing support and feature development.

Continental Talent Entry Program (CTEP) March 2022 - June 2022
Systems Engineer Trainee *Queretaro, Mexico*

- Completed requirements, development, and testing for a door-lock and window-control vehicle prototype.

AI Research (In Progress)

- **Master's thesis - IoT cryptography:** Researching hybrid genetic algorithms and simulated annealing for resource-aware selection of cryptographic algorithms and key lengths; Raspberry Pi evaluation is planned.
- **Bio-inspired optimization:** Collaborating on research for sheep-weight estimation, optimizing objective functions to minimize error between measured and estimated weight.

Technical Skills

Programming & Data: Python, C++, NumPy, Pandas, SQLite
Machine Learning (Coursework): scikit-learn, PyTorch, prediction, neural networks
Optimization (Research): Objective-function optimization, genetic algorithms, simulated annealing, bio-inspired metaheuristics
Evaluation & Delivery: Ground-truth metrics, Dash/Plotly, OpenCV, GitHub Actions, Git, Linux, eCAL
Spoken Languages: English, Spanish

Education

Master of Science in Computer Science (Artificial Intelligence) Expected July 2027
Autonomous University of Queretaro *Online, 2025 - Present*

Bachelor of Science in Engineering Physics 2017 - 2021
Autonomous University of Queretaro *Queretaro, Mexico*